Unsupervised Learning with Agricultural Data

Find natural groups of farm fields that nobody has labelled, and show the groups are real

Unsupervised learning **Field and sensor records** **Uses Weeks 1, 5, 7 and 8**

The task. A farm advisory service has measurements from hundreds of fields: the soil, how wet the ground is through the season, how green the crop looks in satellite images, and how each field is farmed. Nobody has sorted the fields into types. Can you find natural groups of fields that an adviser could treat the same way, and explain what makes each group different?

THE DATA

In Week 9, the teaching team gives you a table of field records with no labels. It has several hundred fields, numbers describing the soil and weather, short series of moisture and crop-greenness readings over the season, and a few columns about how each field is managed. The data is not public, so no AI assistant has seen it. It belongs to the farmers who shared it: use it only for this project.

WHAT YOU WILL DO

1. Explore the data. Decide how to put the columns on comparable scales, and how to sum up each series of readings in a few numbers (Week 1).
2. Use PCA to find the main directions in which fields differ, and give each one a plain name, such as “a wetness direction” (Weeks 5 and 8).
3. Group the fields with K-means and at least one other method. Choose the number of groups by argument, using the elbow and silhouette plots as evidence (Week 5).
4. Train a small autoencoder, a network that learns to squeeze the data down and rebuild it, and compare its groups with PCA’s. Estimate how many numbers you really need to describe a field (Weeks 7 and 8).

WHAT YOU CODE YOURSELF

PCA via SVD is the heart of this project: write it in NumPy and check it against scikit-learn. You also write the autoencoder’s **backpropagation** and **optimiser step** yourself, and check them against PyTorch.

IS IT FAIR AND SAFE?

Advice based on these groups could help some farms more than others. Check whether your groups simply follow farm size or region, and whether farm size can be predicted back from your PCA results (the Week 5 test). Try one fix, such as removing the columns that carry it, and measure what changes. Explain each group with a profile of its typical values. Write a short note on who owns the data and what the farmers agreed to.

QUESTIONS TO PREPARE FOR

Why this number of groups, and what would make you change your mind? What does your first PCA direction mean on a real farm? Are your groups really in the fields, or did your choice of scaling create them?

COMMON MISTAKES

Columns measured in large units drowning out the others in PCA. A silhouette score that prefers two groups for a dull reason. Groups that disappear when you change the random seed.

WEEK-BY-WEEK PLAN

Week 9: data explored and scaled; readings summed up; PCA via SVD tested. **Week 10:** number of groups chosen and argued; main PCA directions named. **Week 11:** autoencoder trained and compared; fairness check with one fix; each group explained. **By 6 January 2027:** report and repository submitted. **Week of 11 January:** 10-minute in-person interview.

TO AIM HIGHER

Show that your groups stay the same when you resample the data or change the seed. Add a two-dimensional map of the fields (t-SNE or UMAP) and say honestly what it distorts. Only after grouping, check whether the groups differ in yield. Write one paragraph of advice for a farmer, in everyday words.